



What did Santa's wife say during the storm?



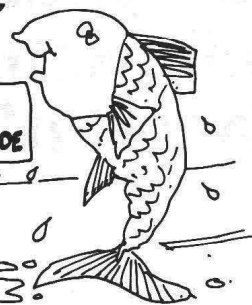
Simplify the expressions given, leaving your answer in index form. Each answer and letter beside the question gives the puzzle code.

b^8	a^{11}	a^8	$18a^6b^5$	e^6	a^7	$10b^5$	$12a^4b^9$	$18a^6b^5$	$18a^6b^5$	
a^7b^7	$6b^{14}$	$18a^6b^5$	$15e^4$	e^6	$6e^7$	a^7	$10b^5$	$18a^6b^5$	e^6	$15e^4$

$e \times e \times e \times e \times e \times e$	$a^6 \times a^5$
a =	= o
$b \times b^3 \times b^4$	$a^3b \times a^4b^6$
c =	= t
$2e^5 \times 3e^2$	$a^4 \times a \times a^2$
i =	= n
$5b^3 \times 2b \times b$	$2b^4 \times 3a^5 \times 3ab$
d =	= e
$a \times a \times a^3 \times a \times a^2$	$3e^2 \times 5e \times e$
m =	= r
$2b^5 \times b^6 \times 3b^3$	$2ab \times 3a^3 \times 2b^8$
h =	= s

Why did the fish cross the road?

SIMPLIFY THE EXPRESSIONS GIVEN, LEAVING YOUR ANSWERS IN INDEX FORM TO DISCOVER THE PUZZLE CODE

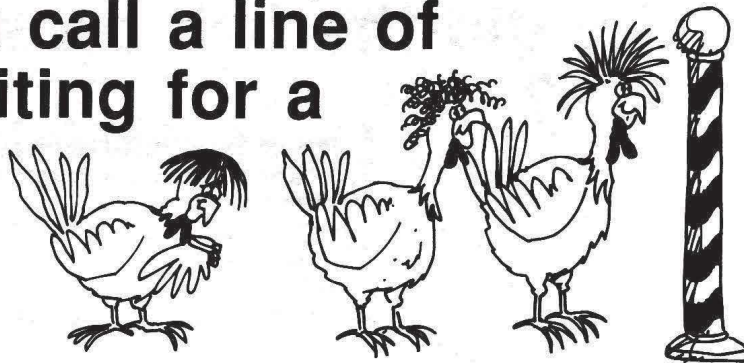


A $8a^6 \div a^4$ =	N $3ab^3 \div a$ =
D $6c^4 \div 3c$ =	O $\frac{10a^{10}}{2a^3}$ =
E $\frac{18e^3f^2}{9e^2f}$ =	R $b^{16} \div b^7$ =
G $\frac{6a^{12}}{3a^8} \times 2a^5$ =	T $\frac{32e^7f^6}{8e^5f^5}$ =
H $24c^{18} \div 4c^7$ =	W $\frac{c \times c \times c \times c \times c}{c \times c \times c}$ =
I $\frac{36b^8 \times b^3}{9b^5 \times 2b^4}$ =	

6c ¹¹	2ef	c ²	8a ²	3b ³	4e ² f	2ef	2c ³	4e ² f	5a ⁷	4a ⁹	2ef	4e ² f	
4e ² f	5a ⁷	4e ² f	6c ¹¹	2ef	5a ⁷	4e ² f	6c ¹¹	2ef	b ⁹	4e ² f	2b ²	2c ³	2ef



What do you call a line of chickens waiting for a haircut?



Join the dot beside each question to the dot next to its answer to find the code.

$5x^\circ =$

$12 - 9b^\circ =$

$a^\circ c^2 b^\circ =$

$21e^3 \div 3e^3 =$

$15 - 7y^\circ + 1 =$

$\frac{24mn^2p}{3mnp} =$

$4a^\circ + 7b^\circ =$

$\frac{-8e^\circ}{-4u^\circ} =$

$12f^7a \div 2f^5a =$

$2a^\circ + 3b^\circ - 4c^\circ =$

$8x^5f \div 2x^5 =$

• 3

• 2

• 7

• 1

• $8n$

• 5

• $6f^2$

• 11

• c^2

• $4f$

• 9

K

5

3

7

N

A

E

R

1

9

H

Q

4

I

U

11

8

6

10

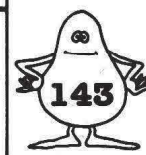
2

B

C

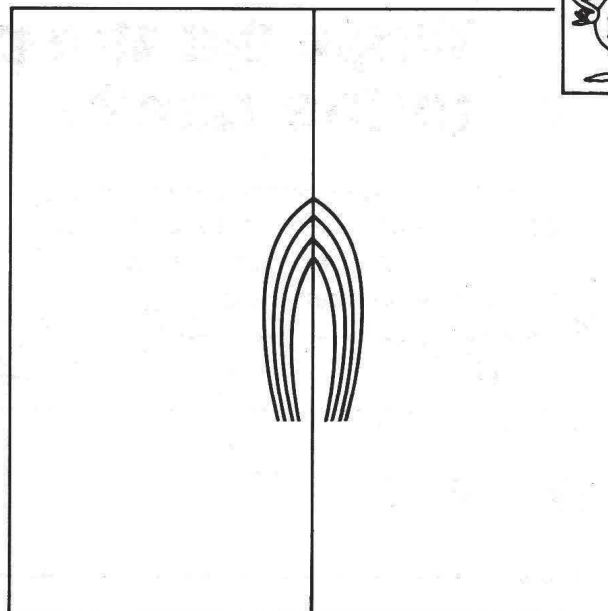
1	2	3	4	2	5	6	7	8	1	9	8	6	9	10	11	6	11	6	!
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Find the title!



EXPAND THE BRACKETS FOUND BELOW.
JOIN THE DOT NEXT TO EACH QUESTION
TO THE DOT NEXT TO ITS ANSWER FOUND
SOMEWHERE ON THE PAGE, EACH LINE
PASSES THROUGH A NUMBER AND LETTER
GIVING THE PUZZLE ANSWER CODE.

1	2	3	4	5	4	6	7	8	9	10	11
12	4	6	12	13	14	5	15	2	2	3	



$(e^3)^4$ • $5(b^1)^6$ • $(c^7)^3$ • $2(e^2)^3$ • $(a^7)^2$
 H D • $4e^{10}$ E • $8b^{12}$
 $(a^5)^3$ 10 I • a^{15} 3
 15 • $5b^6$ 1 K • b^9 11 N
 • $4a^4$ 5 S • 1 • L • $(2e^5)^2$
 2 • $25c^4$ C • 2
 $2(c^0)^{15}$ 8 • e^{16} 14 6
 • $2e^6$ 9 U • $(a^3)^0$
 $(b^5)^4$ 13 R • b^{20}
 0 7 4 12 • e^{12} B
 • c^{21} A • a^{14} T
 $(e^2)^8$ • $(2a^2)^2$ • $(b^3)^3$ • $(5c^2)^2$ • $(2b^4)^3$



What did the puddle say to the rain?

Simplify these terms and write the answers in index form to find the puzzle answer code.

$$\mathbf{D} = \left(\frac{2x}{y}\right)^2 =$$

$$\mathbf{N} = \left(\frac{y}{2}\right)^4 =$$

$$\mathbf{E} = \left(\frac{x^2}{y}\right)^3 =$$

$$\mathbf{O} = \left(\frac{y^2}{x}\right)^3 =$$

$$\mathbf{E} = \left(\frac{x}{2}\right)^3 =$$

$$\mathbf{O} = \left(\frac{x}{3}\right)^3 =$$

$$\mathbf{I} = \left(\frac{y^3}{3}\right)^2 =$$

$$\mathbf{P} = \left(\frac{2y}{3}\right)^2 =$$

$$\mathbf{I} = \left(\frac{y}{2x}\right)^2 =$$

$$\mathbf{R} = \left(\frac{5}{xy}\right)^2 =$$

$$\mathbf{M} = \left(\frac{xy^2}{2}\right)^2 =$$

$$\mathbf{S} = \left(\frac{x^3y}{2}\right)^4 =$$

$$\mathbf{M} = \left(\frac{1}{x^2}\right)^4 =$$

$$\mathbf{T} = \left(\frac{3}{x^2}\right)^3 =$$



$\frac{4x^2}{y^2}$	$\frac{25}{x^2y^2}$	$\frac{y^6}{x^3}$	$\frac{4y^2}{9}$	$\frac{y^6}{9}$	$\frac{y^4}{16}$	$\frac{x^{12}y^4}{16}$	$\frac{x^3}{27}$	$\frac{x^2y^4}{4}$	$\frac{x^6}{y^3}$	$\frac{27}{x^6}$	$\frac{y^2}{4x^2}$	$\frac{1}{x^8}$	$\frac{x^3}{8}$
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